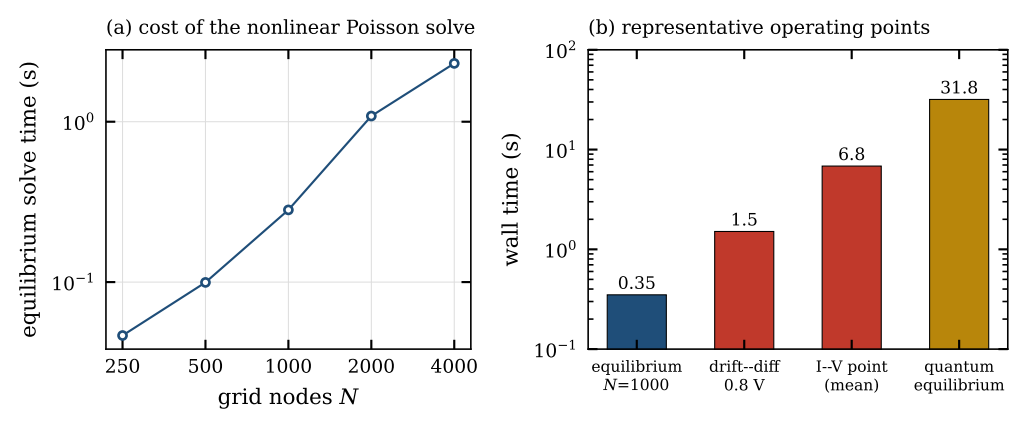
(a) cost of the nonlinear Poisson solve
equilibrium solve time (s)
10^{0}
10^{-1}
250
500
1000
2000
4000
grid nodes N
(b) representative operating points
wall time (s)
10^{2}
10^{1}
10^{0}
10^{-1}
0.35
1.5
6.8
31.8
equilibrium
N=1000
drift--diff
0.8 V
I–V point
(mean)
quantum
equilibrium